

## Today's Agenda:-

1. Transpose of a Square Matrix.
2.  $90^\circ$  rotation of a square matrix
3. Maximum Number of 1's  
→ 2 variations
4. Reverse string word by word.

1. Transpose of a Square Matrix  
rows = columns  
↓  
rows & columns get interchanged.

in-place

|   |   |   |   |
|---|---|---|---|
|   | 0 | 1 | 2 |
| 0 | 1 | 2 | 3 |
| 1 | 4 | 5 | 6 |
| 2 | 7 | 8 | 9 |

input

|   |   |   |   |
|---|---|---|---|
|   | 0 | 1 | 2 |
| 0 | 1 | 4 | 7 |
| 1 | 2 | 5 | 8 |
| 2 | 3 | 6 | 9 |

O/P:

Observation:-

1. Principal Diagonal remains same
2. Every element at  $\text{idx } (i, j)$  has been swapped with element at  $\text{idx } (j, i)$

$3 \times 4 \longrightarrow 4 \times 3$

Approach:- Iterate on either of the  
2 halves & swap  $(i, j)$  with  $(j, i)$ .

# Code.

```
for (i = 0; i < N; i++) {  
    for (j = i + 1; j < N; j++) {  
        swap(arr[i][j],  
              arr[j][i]);  
    }  
}
```

↑  
temp = arr[i][j];  
arr[i][j] = arr[j][i];  
arr[j][i] = temp;

Q:-1

T.C  $\rightarrow O(N^2)$

S.C  $\rightarrow \underline{O(1)}$ .

If not square:

```
int [][] trans = new int [m][N];
```

```
for (row = 0; row < m; row++) {
```

```
    for (col = 0; col < N; col++) {
```

```
        trans [col][row] = mat [row][col];
```

3

3

```
    return trans;
```

2. Rotate a sq matrix  $90^\circ$  clockwise.

input

|   | 0 | 1 | 2 |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
| 1 | 4 | 5 | 6 |
| 2 | 7 | 8 | 9 |

$90^\circ$   
clockwise  
rotation.

in-place

|  | 0 | 1 | 2 |   |
|--|---|---|---|---|
|  | 3 | 5 | 1 | 0 |
|  | 6 | 4 | 2 | 1 |
|  | 9 | 8 | 7 | 2 |

transpose

|   | 0 | 1 | 2 |
|---|---|---|---|
| 0 | 1 | 4 | 7 |
| 1 | 2 | 5 | 8 |
| 2 | 3 | 6 | 9 |

swap

each row.

7:43

```
int[][] rotate 90 (int[][] mat) {
```

```
    mat = transpose(mat);
```

```
    for (row = 0; row < N; row++) {
```

```
        reverse(mat[row]);
```

```
    }
```

```
    return mat;
```

```
}
```

T.C  $\rightarrow O(N^2)$

S.C  $\rightarrow O(1)$

3. Maximum no' of consecutive 1's after replacement.

Given an array of 1's & 0's, you are allowed to replace only one 0 with 1. Find max. cont ones that can be obtained after making the replacement.

ex:- { 1, 1, 0, 1, 1, 0, 1, 1, 1 }

ans = 5

Q:- 2 { 1, 1, 0, 1, 1, 0, 1, 1, 1 }

ans = 6

$$Q \div 3$$

ans = 6 ∴

Idea:-

Iterate on the array. For every zero, replace it with a 1 & find the max sequence that is getting generated.

Col. consecutive ones on left = l  
" " " " " right = r

max Sequence Hct Ts getting  
generated is  $(k+1)$

compare with global ans.

$\{1, 1, 0, 1, 1, 0, 1, 1, 1\}$

ans = ~~156~~

$$\gamma = 3$$
$$l = 2$$

$\{1, 1, 1, 1, 1\}$ ; ans = 5;



# # Code

```
int solve (int[] arr) {
```

```
    int totalOnes = 0;
    for (i=0; i < N; i++) {
        if (arr[i] == 1) { totalOnes++; }
    }
    if (totalOnes == arr.size()) {
        return N;
    }
```

```
    int ans = 1;
    for (i=0; i < N; i++) {
        if (arr[i] == 0) {
            l = 0; r = 0;
            j = i-1;
            while (j >= 0 && arr[j] == 1) {
                j--; l++;
            }
            j = i+1;
            while (j < N && arr[j] == 1) {
                j++; r++;
            }
            ans = max(ans, l+r+1);
        }
```

3  
3

return ans;

3

Q:- 4

{ 1, 1, 0, 1, 1, 0, 1, 1, 1 }  
i  
j

0 - - - 1 - - - 0

At max any element is  
being accessed 3 times.

$\therefore$  Total iterations =  $3N$ .

$$\therefore \underline{T_c = O(N)}$$

## Variation:-

Maximum no' of consecutive 1's  
after swapping.

Given an array of 1's & 0's, you are allowed to swap only one 0 with 1. Find max. cont ones. that can be obtained after making the swapping.

ex: { 1, 0, 1, 1, 0, 1 }

ans = 4.

Q:- 5

{ 1, 1, 0, 1, 1, 1 }

ans = 5.

8:38

## # Code

```
int totalOnes = 0;
for (i=0; i < N; i++) {
    if (arr[i] == 1) { totalOnes++; }
}
if (totalOnes == arr.size()) {
    return N;
}
```

```
int ans = 1;
for (i=0; i < N; i++) {
    if (arr[i] == 0) {
        l = 0; r = 0;
        j = i-1;
        while (j >= 0 && arr[j] == 1) {
            j--; l++;
        }
        j = i+1;
        while (j < N && arr[j] == 1) {
            j++; r++;
        }
        if (l+r+1 <= totalOnes) {
            ans = max(ans, l+r+1);
        }
        else {
            ans = max(ans, l+r+1);
        }
    }
}
```

$\text{return (lto);}$   
 $\text{return ans;}$

8:50

$i$   
 $\{1, 0, 1, 1, 0, 1\}$

Count Zero = 0

$l$     $r$   
 $0$     $0$

$i$   
 $\{0, 0, 0, 0, 0\}$   
 $j$

✓

totalans = 0

$i$   
 $\{1, 1, 1, 1, 1\}$

4. Reverse string word by word.

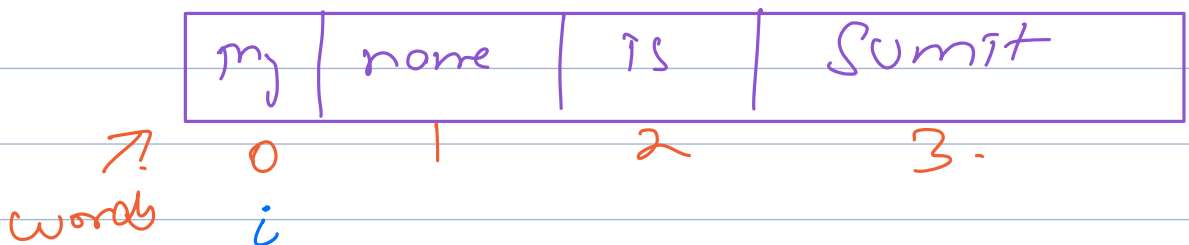
String S = "my name is Sumit";

o/p:- "Sumit is name my";

Approach:

① Split the string of words  
into array of words.

String[] words = S.split(" ");



String Builder ans = "Sumit is name  
my";

return ans;

Code till 0:05

```

class Solution {
    public String reverseWords(String s) {
        A = A.trim();
        String[] words = A.split("\\s+");
        StringBuilder sb = new StringBuilder();
        for (int i = words.length - 1; i >= 0; i--) {
            sb.append(words[i]);
            if (i > 0) {
                sb.append(" ");
            }
        }
        return sb.toString();
    }
}

```

Python

```

class Solution:
    def solve(self, A):
        A = A.strip()
        A = A.split()
        A = A[::-1]
        return ' '.join(A)

```

7 am → 4 Questions

75+ marks

3/4 Questions



$$> 10^7 - 10^8$$

---

1, 4, 1 mSET ; 3 Reattempts.